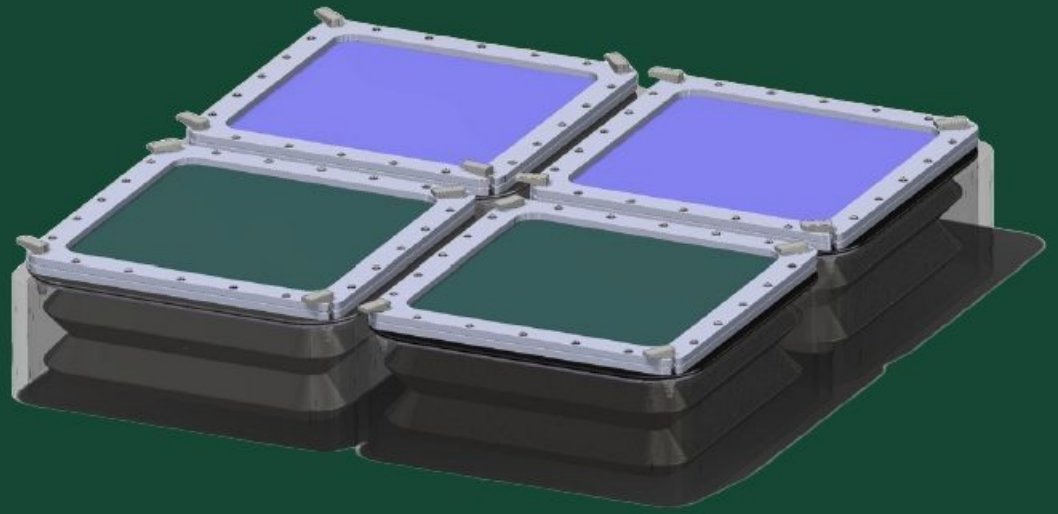
# MASS

## Modular Acoustic Suppression System



**Student Team**: Julia Megargle, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Michael Milner, Emmi Cayer

**Advisors**: Dr. John Chen, Rick Lasko

# Problem Statement

**Current acoustic environments (ISS) can exceed noise limits and create risk to the crew**

## The Challenge

### Acoustic standards exist to:

- Prevent hearing loss
- Maintain communications
- Support crew performance [1]

### ISS Monitoring Results

ISS acoustic monitoring has identified locations and operating conditions that exceed NASA noise requirements. [2]

### Crew Exposure

41% of crew dosimetry measurements have exceeded NASA noise exposure limits since 2008. [3]

## Impacts on Crew

### Communication

- Degraded vocal communication
- Reduced ability to hear critical audio cues

### Performance

- Reduced concentration
- Reduced productivity

### Habitability

- Sleep disruption
- Reduced crew comfort

### Health

- Temporary or permanent hearing loss [1]
- Irritation and headaches [6]



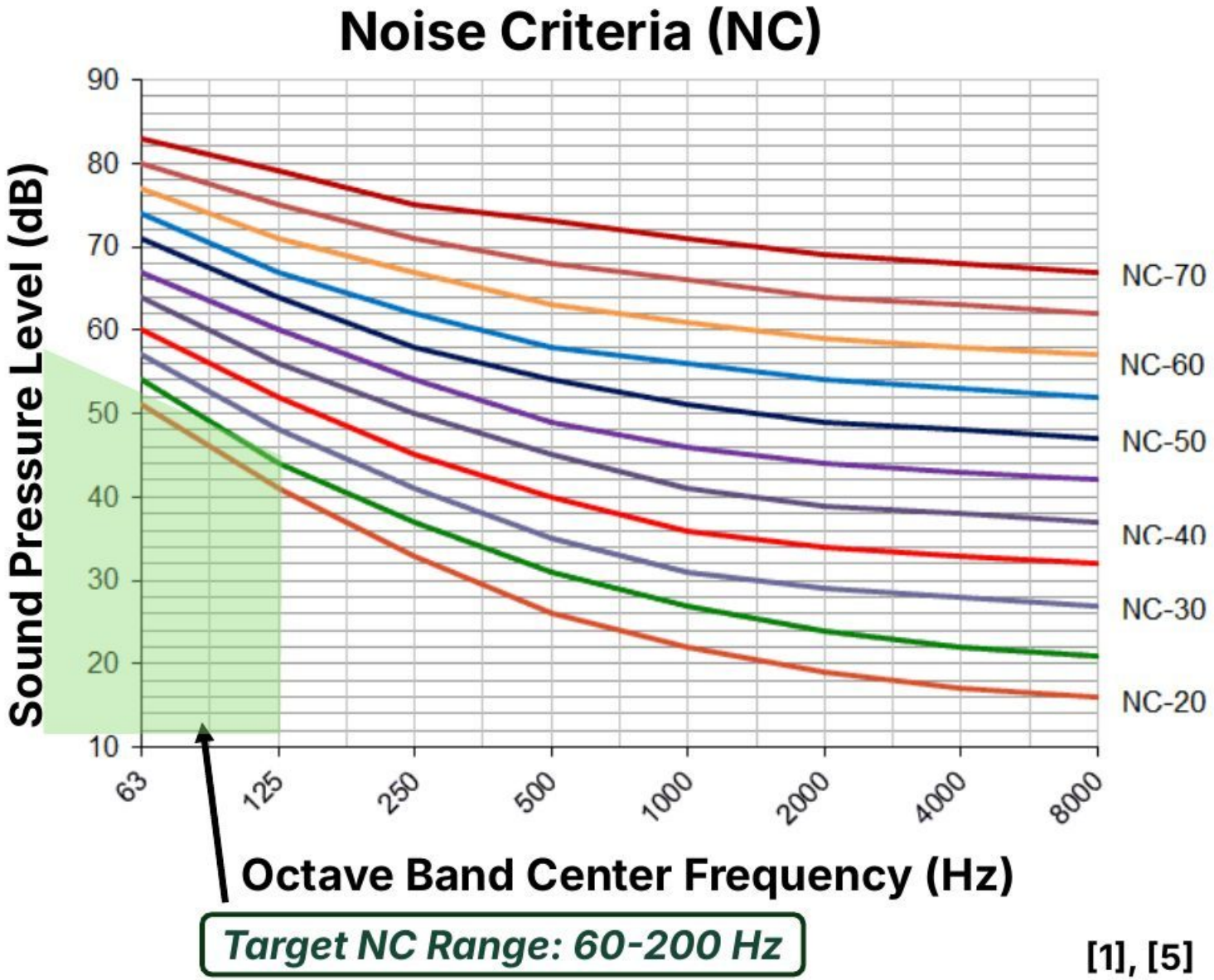
# Objectives

**Meet NC-40**
Meet NASA NC-40 acoustic requirements

**Suppress Dominant ECLSS Frequencies**
Range of 60-200 Hz

**Maintain Low Power**
Maintain power consumption below 1.5 kW

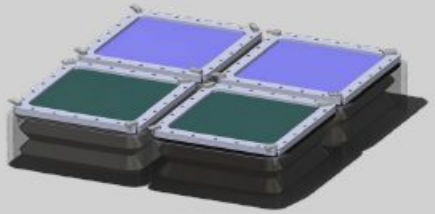
**Minimize Mass & Volume**
Remain below 100 lb system mass and fit within limited cabin volume




[1], [5]

# Existing Solutions

**Current acoustic treatments are generally passive, bulky, and not optimized for adaptive low-frequency suppression.**

| Solution | Visual Example | Adaptive Response | Compact | Addresses Low Frequency |
|---|---|---|---|---|
| Acoustic Blankets |  | ✗ | | ✗ |
| Acoustic Panels |  | ✗ | ✗ | ✗ |
| Vibration Isolators |   | | | ✗ |
| MASS |  | ✓ | ✓ | ✓ |

[1], [4]

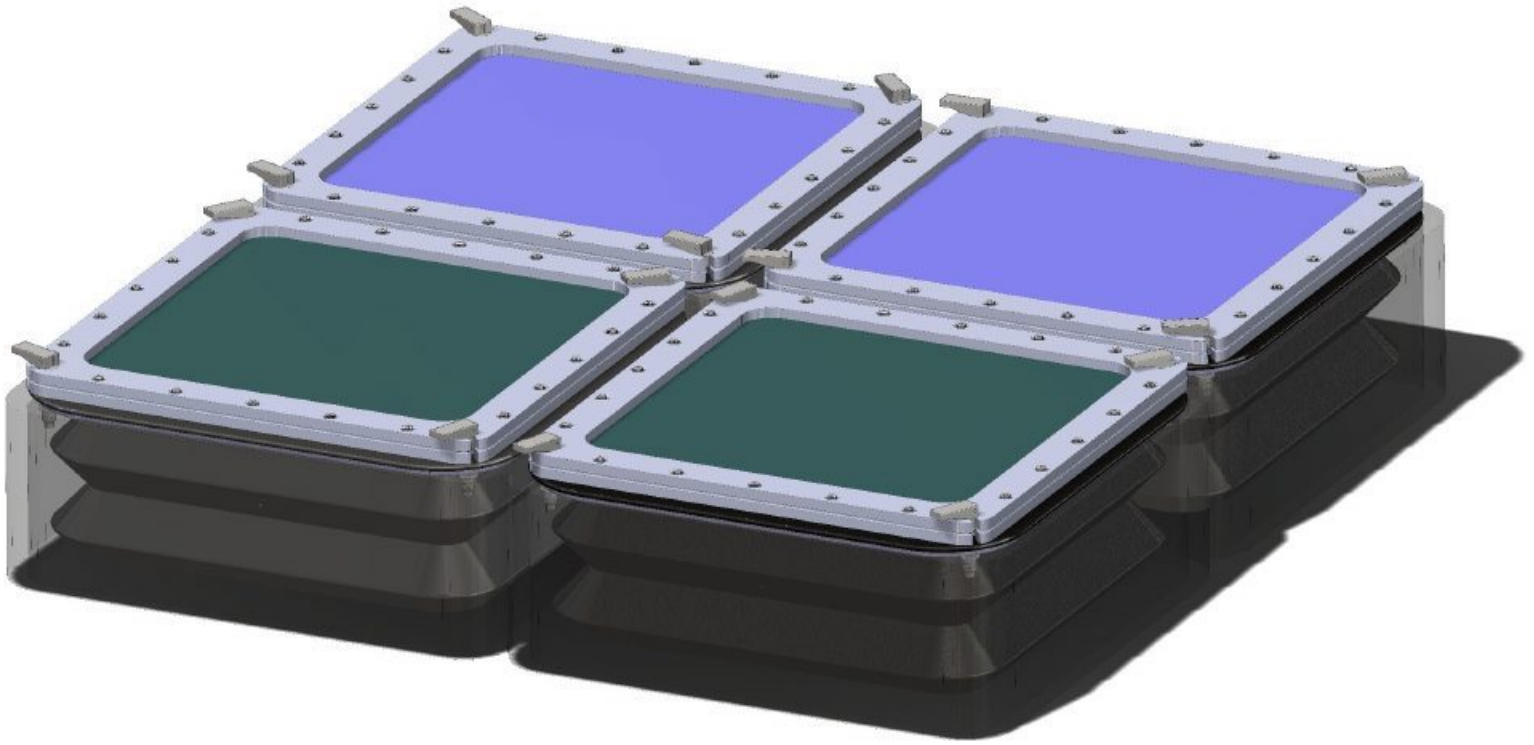
# MASS Overview

*Adaptive tuned membrane absorber for low-frequency ECLSS noise*

**60-200 Hz**
Targets the dominant ECLSS noise band

**Adaptive**
Automatically retunes to changing frequencies

**Modular**
Scalable panel architecture for flexible integration



[4]

# How MASS Works

**1 Noise Source**



ECLSS equipment produces acoustic noise within the cabin

**2 Microphone**



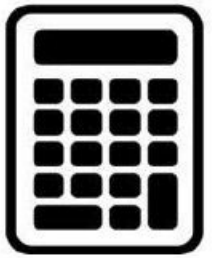
Microphone sensors detect acoustic noise

**3 Frequency Detection**



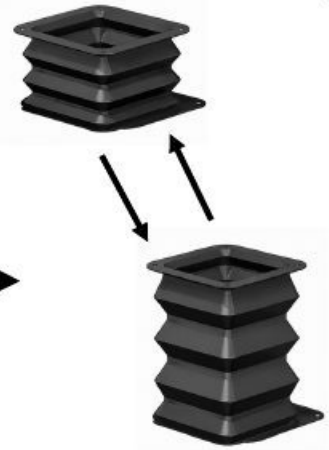
Signal processing identifies dominant frequencies

**4 Depth Calculation**



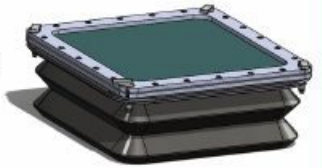
Controller calculates required cavity depth

**5 Actuator Motion**



Actuators move the mounting plate to required cavity depth

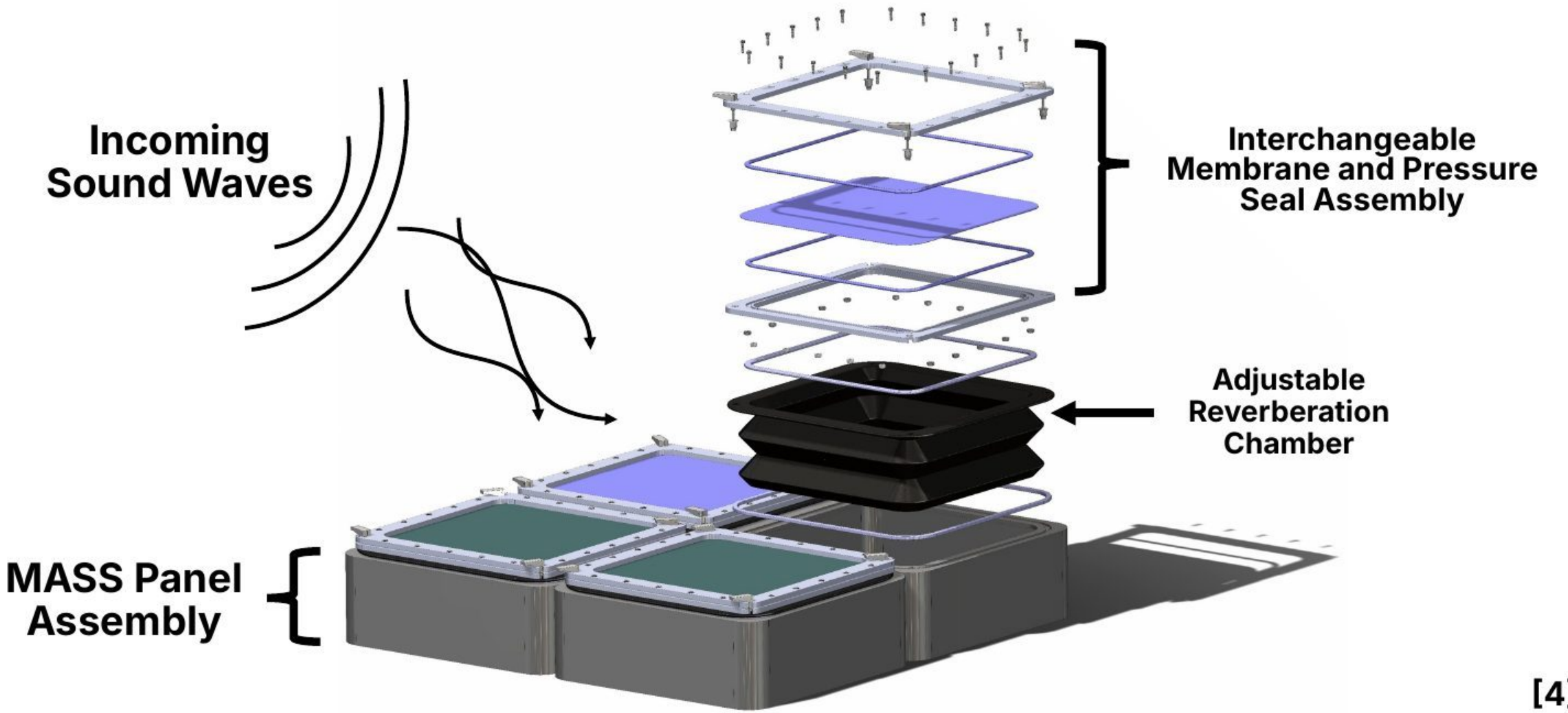
**6 Noise Suppression**



Coupled membrane-cavity system resonates, dissipating acoustic energy

**[4]**

# MASS System - Components




[4], [7]

# Membrane Absorber Model

*Mass behaves like a spring-mass system where cavity depth controls the tuned frequency.*

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0}{m_s d}}$$

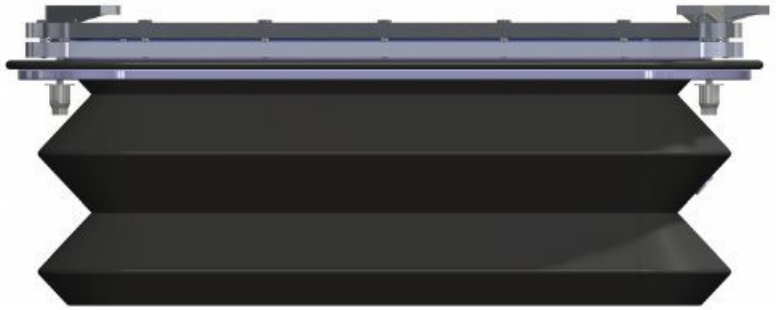
***Key parameters:***

$m_s$ - *membrane surface density*

$d$ - *cavity depth*

**LARGER CAVITY DEPTH**

**Greater Depth**



**LOWER FREQUENCY**

**60 Hz**

**SMALLER CAVITY DEPTH**



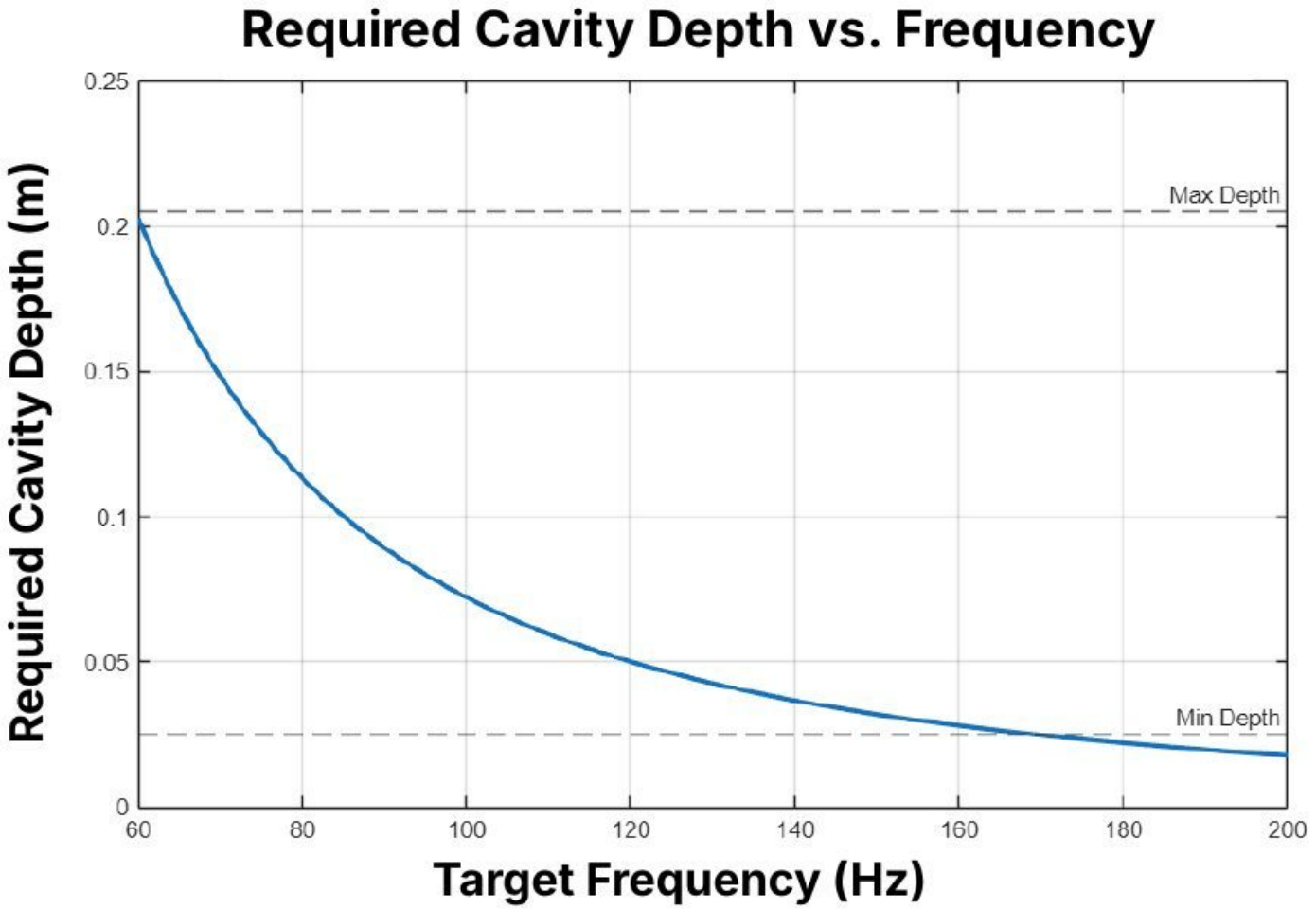
**Less Depth**

**HIGHER FREQUENCY**

**200 Hz**

[4]

# Validation – Theoretical Analysis



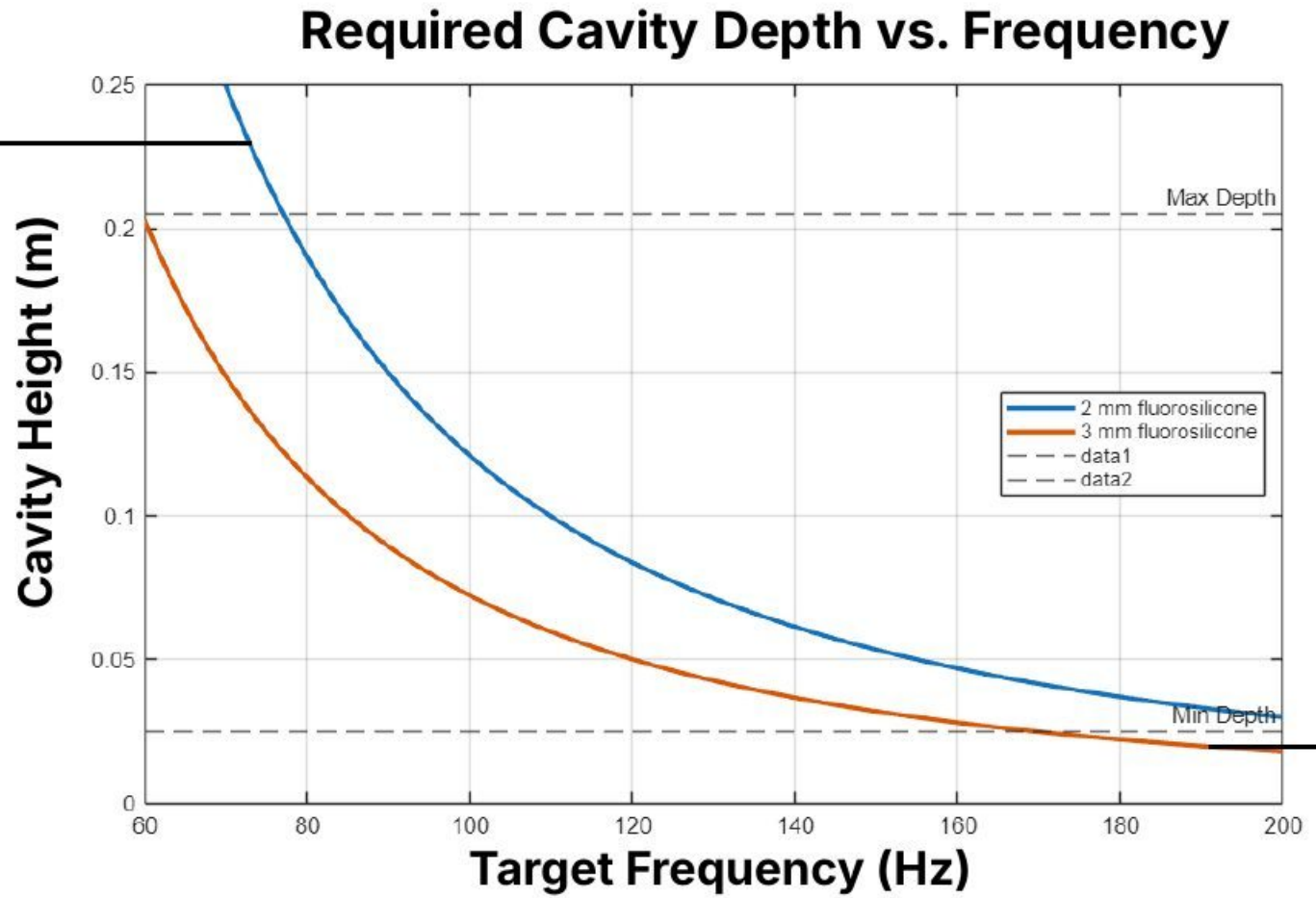

**3 mm fluorosilicone suppresses 60–170 Hz.**

[4]

# Validation – Full Frequency Coverage

2 mm membrane

→ 76–200 Hz




3 mm membrane

→ 60–170 Hz

**Together: Covers entire target range of 63-200 Hz**

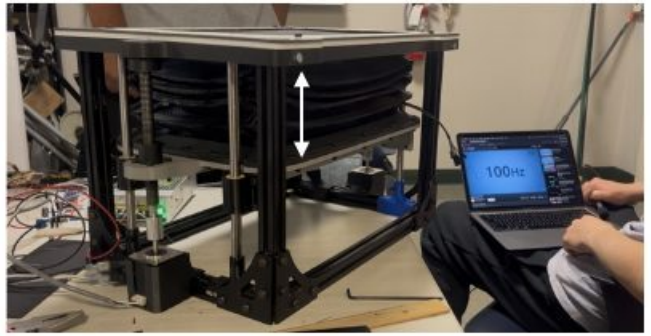
[4]

# Validation – Feasibility Prototype

*Prototype testing verified core functionality, pressure equalization, and structural feasibility of MASS*

## 1. AUTOMATIC TUNING

Actuators tuned cavity depth to target frequencies

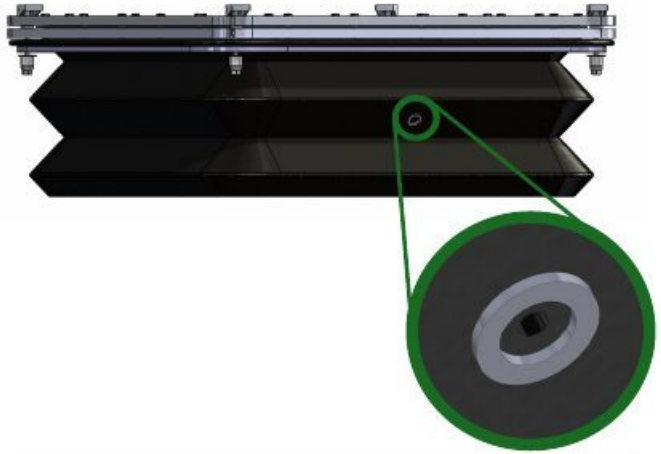



**60-250 Hz**

Automatic tuning verified

## 2. PRESSURE EQUALIZATION

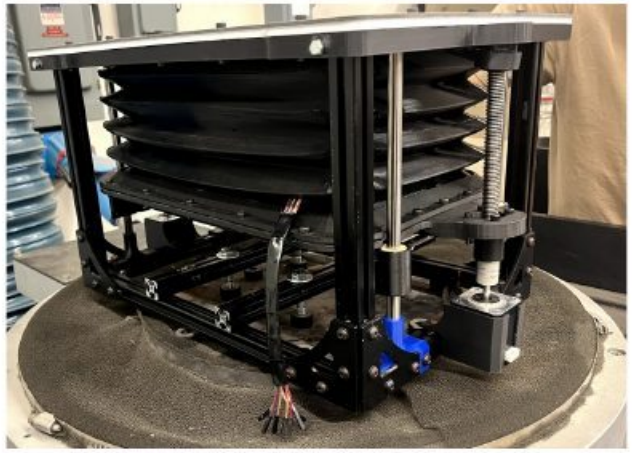
Internal pressure equalized through 13mm aperture



**4.49 sec**

Average equalization time

## 3. STRUCTURAL VERIFICATION



**10.3 g RMS**

Random vibration profile survived

# NASA Integration Requirements

**FREQUENCY RANGE**

**Target**
60-200 Hz

**MASS Value**
60-200 Hz

**MET**

**POWER CONSUMPTION**

**Target**
< 1.5 kW

**MASS Value**
1.36kW

**MET**

**SYSTEM MASS**

**Target**
<100 lb

**MASS Value**
~83.5 lb

**MET**

**MAXIMUM DEPTH**

**Target**
Minimize cabin intrusion

**MASS Value**
~8 in

**MET**

***Mass meets all primary constrains for adoption on the Artemis Human Lander.***

[5], [6]

# Top Risks & Mitigations

| Risk | Consequence | Mitigation |
|---|---|---|
|  **High Pressure Differential** | **Incorrect Tuning Frequency** | • 13 mm pressure equalization aperture<br>• Validated sealing behavior during chamber adjustment |
|  **Electrical System Failure** | **Loss of Adaptive Capability** | • Space-rated wiring and components<br>• Fail-safe mode holds chamber at default ECLSS frequency |
|  **Actuator Position Error** | **Incorrect Cavity Depth** | • Position feedback from actuators<br>• Calibration checks Software travel limits |

***Key risks have been identified, and mitigation strategies have been incorporated into the MASS design***

[4]

# Budget and Cost Estimate

**$5.63M**

**Total Development Cost**

**3- year Development Program**

**30% Contingency Margin Included**

| Major Cost Driver | Cost |
|---|---|
| Personnel | $3.78M |
| Testing & Facilities | $0.25M |
| Hardware & Fabrication | $0.30M |
| Cost Margin | $1.30M |
| **Total** | **$5.63M** |

## Key Assumptions

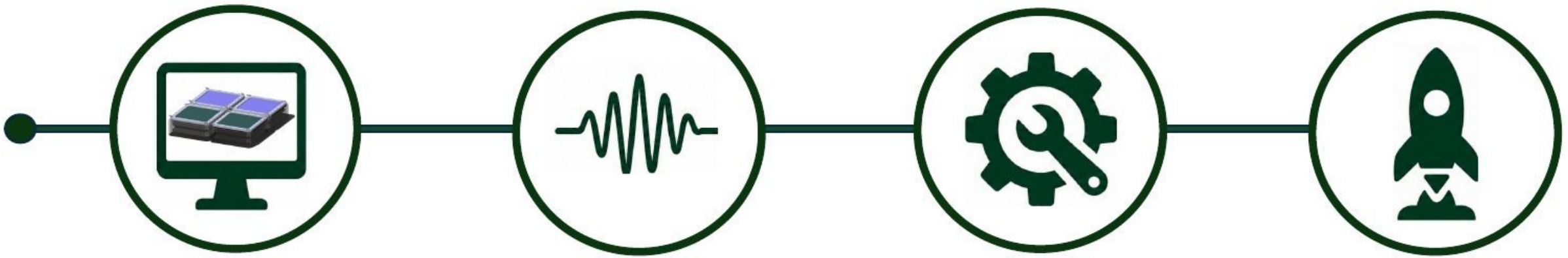
- 10- person engineering team
- 3-year development schedule
- One flight-ready MASS unit
- 30% contingency margin

Cost estimate developed using a NASA-style lifecycle approach from prototype development through flight implementation.

[5]

# Path to Flight

*A phased approach to design, validate, and implement the MASS for Artemis Missions*



## 2026 — Design Refinement

- Material Selection
- CAD refinement
- Prototype updates

TRL 3-5

## 2027 — Testing & Validation

- Anechoic chamber insertion-loss testing
- Lander environment validation
- Design iteration

TRL 6-7

## 2028 — Fabrication

- Final manufacturing
- System Integration

TRL 8

## 2029 — Implementation

- Multi-panel MASS deployment
- Artemis ECLSS integration

TRL 9

[5]

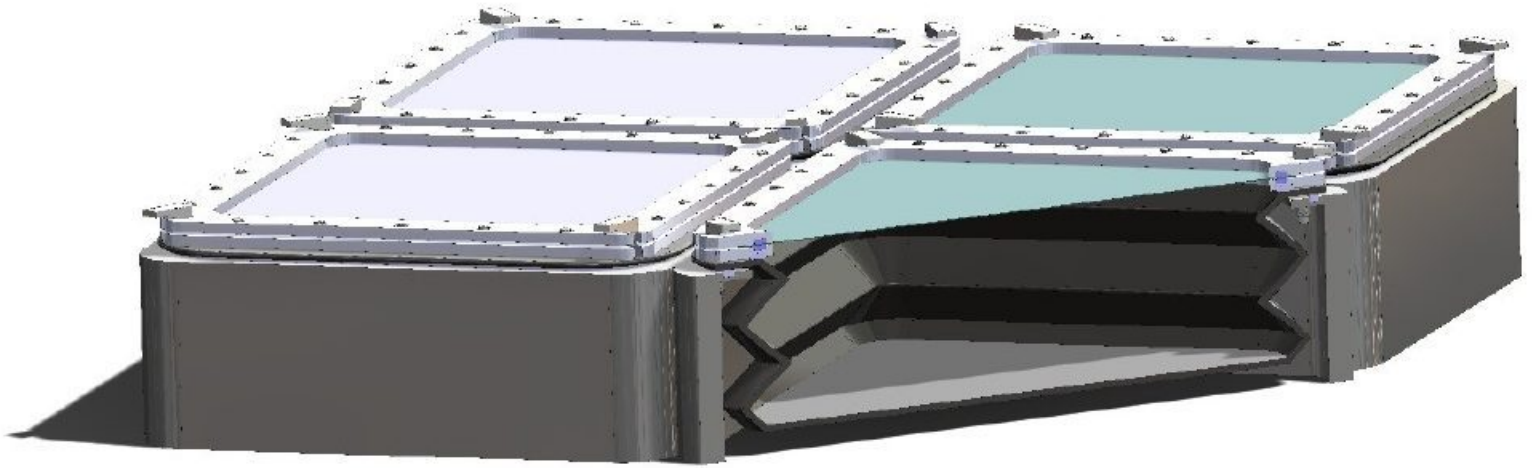
# Why MASS Matters

**PROBLEM**

Current acoustic treatments are passive and struggle with adaptive low-frequency suppression.

**SOLUTION**

MASS is an adaptive tuned membrane absorber that automatically adjusts cavity depth to target dominant ECLSS frequencies.



**VALUE TO NASA**

- ✓ Improves crew habitability and communication
- ✓ Reduces disruptive low-frequency ECLSS noise
- ✓ Meets NASA mass, power, and volume constraints
- ✓ Prototype feasibility demonstrated
- ✓ Supports long-duration Artemis missions

# References


[1] NASA Office of the Chief Health and Medical Officer (OCHMO), Technical Brief TB-035: Acoustics, 2023.

[2] Limero et al., Internal Acoustics of the International Space Station and Other Spacecraft, NASA, 2017.

[3] Limardo et al., Status — International Space Station (ISS) Crewmembers' Noise Exposures, NASA, 2021.

[4] Zhao et al., Theoretical Model of Membrane Acoustic Absorber with Compact Magnet, JASA, 2021.

[5] NASA Human Lander Challenge Competition Flyer, 2026.

[6] NASA-STD-3001 Volume 2, Human Factors, Habitability, and Environmental Health, 2025.

[7] Fluorosilicone in Aerospace: High Performance Sealing, RD Rubber Technology Corp., 2025.

[8] J. R. Goodman and F. W. Grosveld, NASA/SP-2015-624: Acoustics and Noise Control in Space Crew Compartments, 2015.


**[5], [6]**

# Appendices

# Appendix A. Full Membrane Absorber Derivation

When incident sound waves excite the membrane, the membrane behaves like an oscillating mass. The sealed air cavity behind the membrane provides a restoring force due to compression of the trapped air.

**Pressure change in the cavity:**

$$\Delta P = -\gamma P_0 \frac{\Delta V}{V}$$

where:

$\gamma$= ratio of specific heats for air (~1.4)

$P_0$= ambient pressure

$V$= cavity volume

$\Delta V$= change in cavity volume

For membrane displacement $x$ over area $A$:

$$\Delta V = Ax$$

Since cavity volume is:

$$V = Ad$$

where $d$is cavity depth,
substituting gives:

$$\Delta P = -\gamma P_0 \frac{Ax}{Ad}$$

$$\Delta P = -\gamma P_0 \frac{x}{d}$$

This produces an effective acoustic spring constant per unit area:

$$k' = \frac{\gamma P_0}{d}$$





# Appendix A. Full Membrane Absorber Derivation Cont.

The membrane surface mass density $m_s$ acts as the oscillating mass per unit area.

The membrane-cavity system therefore behaves as a simple harmonic oscillator:

$$\omega_0 = \sqrt{\frac{k'}{m_s}}$$

Substituting for $k'$:

$$\omega_0 = \sqrt{\frac{\gamma P_0}{m_s d}}$$

Converting angular frequency to frequency in Hz:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0}{m_s d}}$$

This equation defines the resonance frequency of the membrane absorber.

Rearranging to solve for cavity depth:

$$d = \frac{\gamma P_0}{(2\pi f_0)^2 m_s}$$

This relationship is used by MASS to determine the required cavity depth needed to suppress a measured dominant frequency.

# Appendix A. Full Membrane Absorber Derivation Cont.

**Key Result:**

$$f_0 \propto \frac{1}{\sqrt{d}}$$

Larger cavity depth → lower frequency suppression

Smaller cavity depth → higher frequency suppression

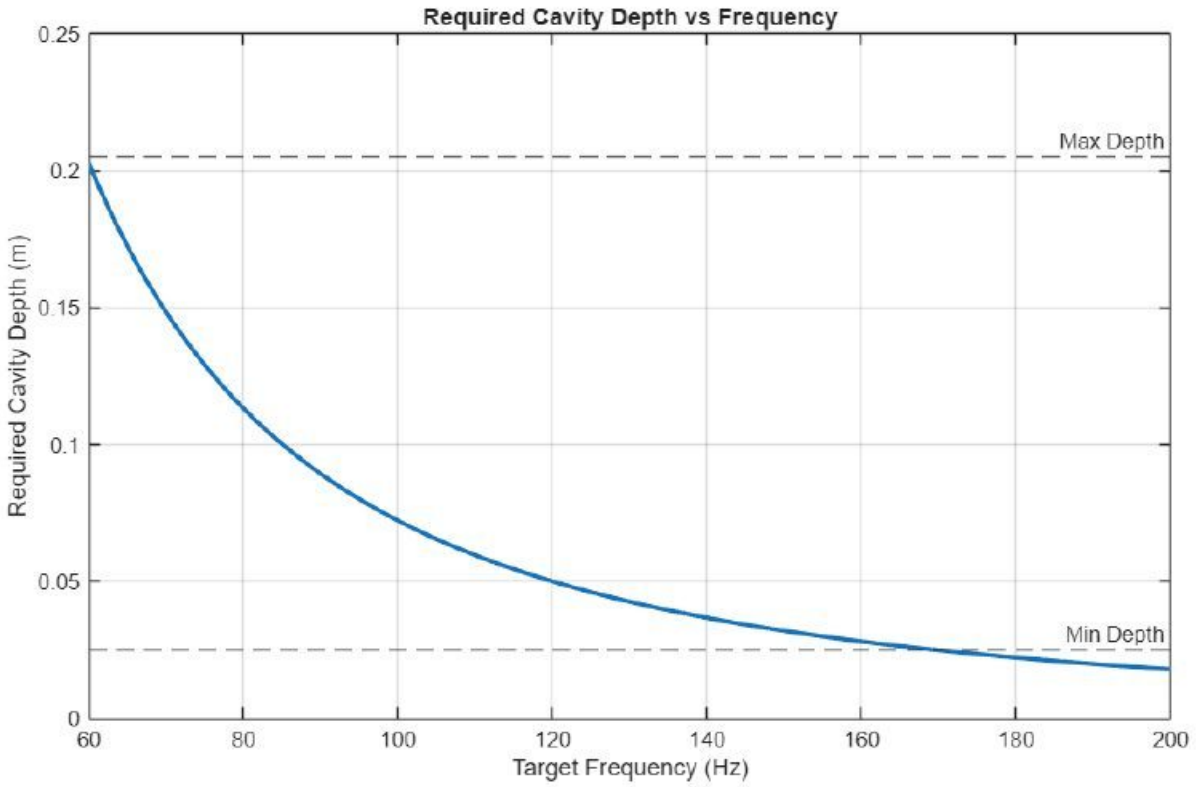
Therefore, adjusting cavity depth allows MASS to tune its resonance frequency and suppress noise across the target 60–200 Hz range.

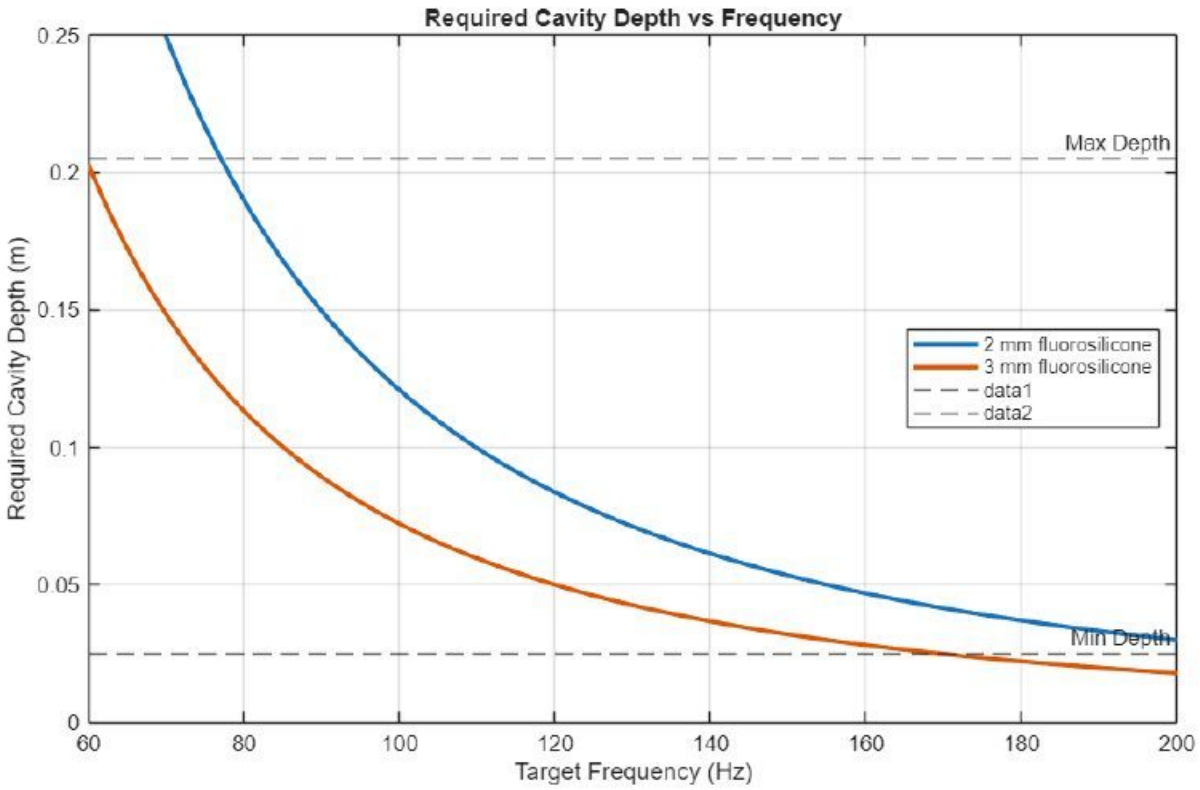
$$\boxed{d = \frac{\gamma P_0}{\left(2\pi f_0\right)^2 m_s}}$$





# Appendix B. Full Frequency-Depth Plots



# Appendix C. Full Pressure Equalization Calculations

To size the pressure equalization aperture, use the orifice flow equation:

$$Q = C_d A \sqrt{\frac{2\Delta P}{\rho}}$$

where:

- $Q$= volumetric flow rate $(m^3/s)$
- $C_d$ =discharge coefficient
- $A$= hole area $(m^2)$
- $\Delta P$= pressure difference $(Pa)$
- $\rho$ =air density $(kg/m^3)$

For a circular hole:

$$A = \frac{\pi d^2}{4}$$

Substitute area into the orifice equation:

$$Q = C_d \left(\frac{\pi d^2}{4}\right) \sqrt{\frac{2\Delta P}{\rho}}$$

Solve for hole diameter:

$$d = \sqrt{\frac{4Q}{\pi C_d \sqrt{2\Delta P/\rho}}}$$

Using the report assumptions:

$$Q = 0.001\, m^3/s$$
$$C_d = 0.6$$
$$\Delta P = 100\, Pa$$
$$\rho = 1.2\, kg/m^3$$

Final result:

$$d \approx 13\, mm$$

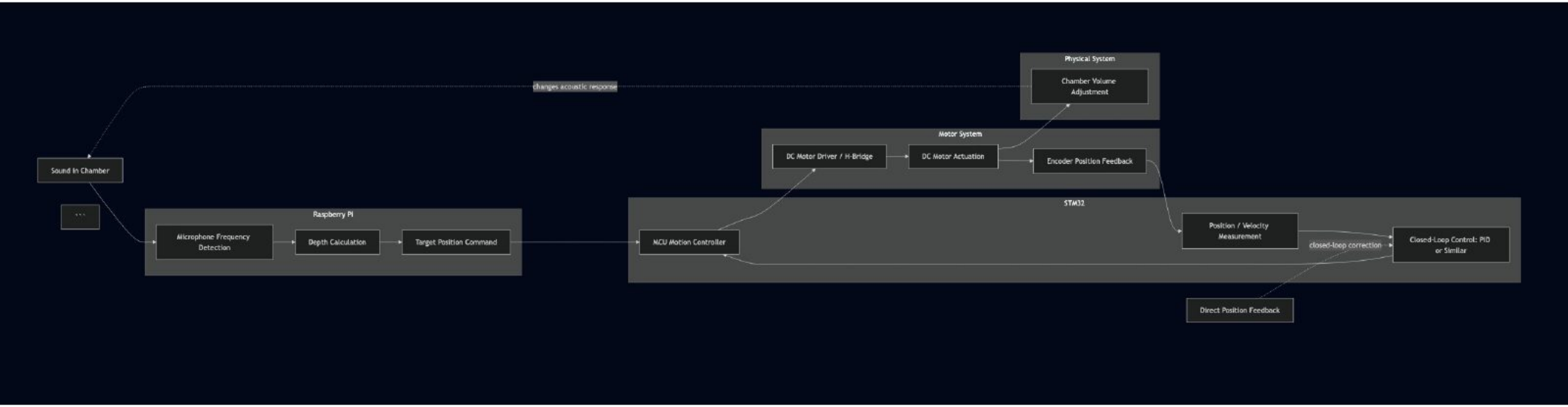
**Conclusion:** A **13 mm aperture** is sufficient for pressure equalization during chamber adjustment.

# Appendix D. Chamber Pressure Equalization

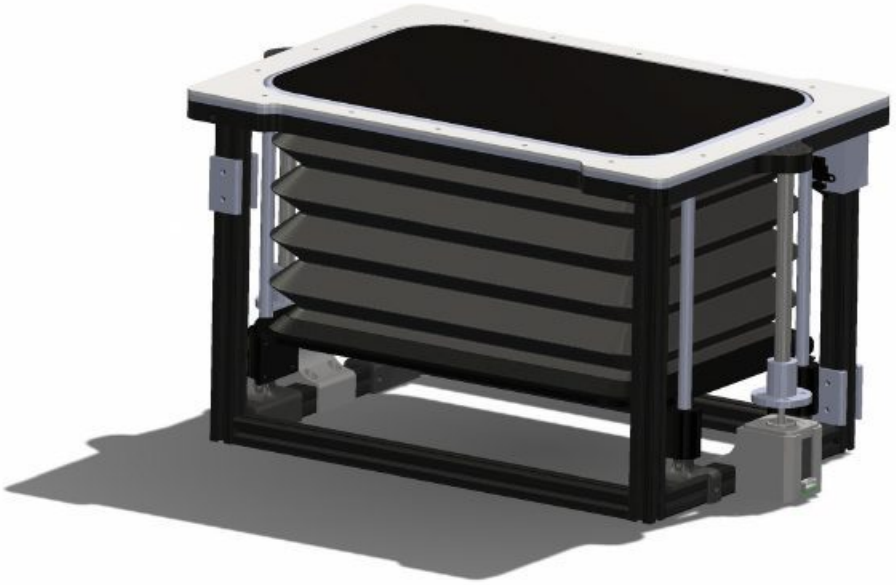
| Actuation Distance [in] | Initial Pressure Difference [kPa] | Time To Equalize [seconds] |
|---|---|---|
| 3 | 1.1 | 3.89 |
| 3 | 1 | 4.86 |
| 3 | 1.2 | 4.68 |
| 3 | 1.3 | 5.34 |
| 3 | 1 | 3.66 |

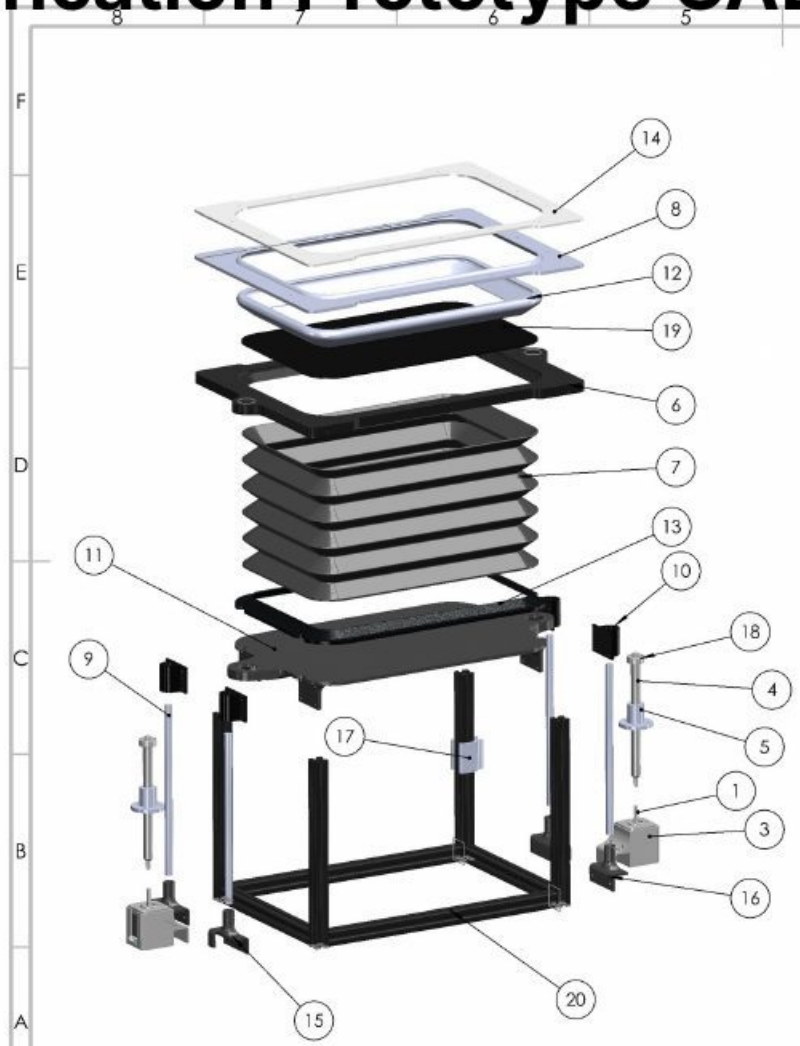


# Appendix E. Actuation Controls



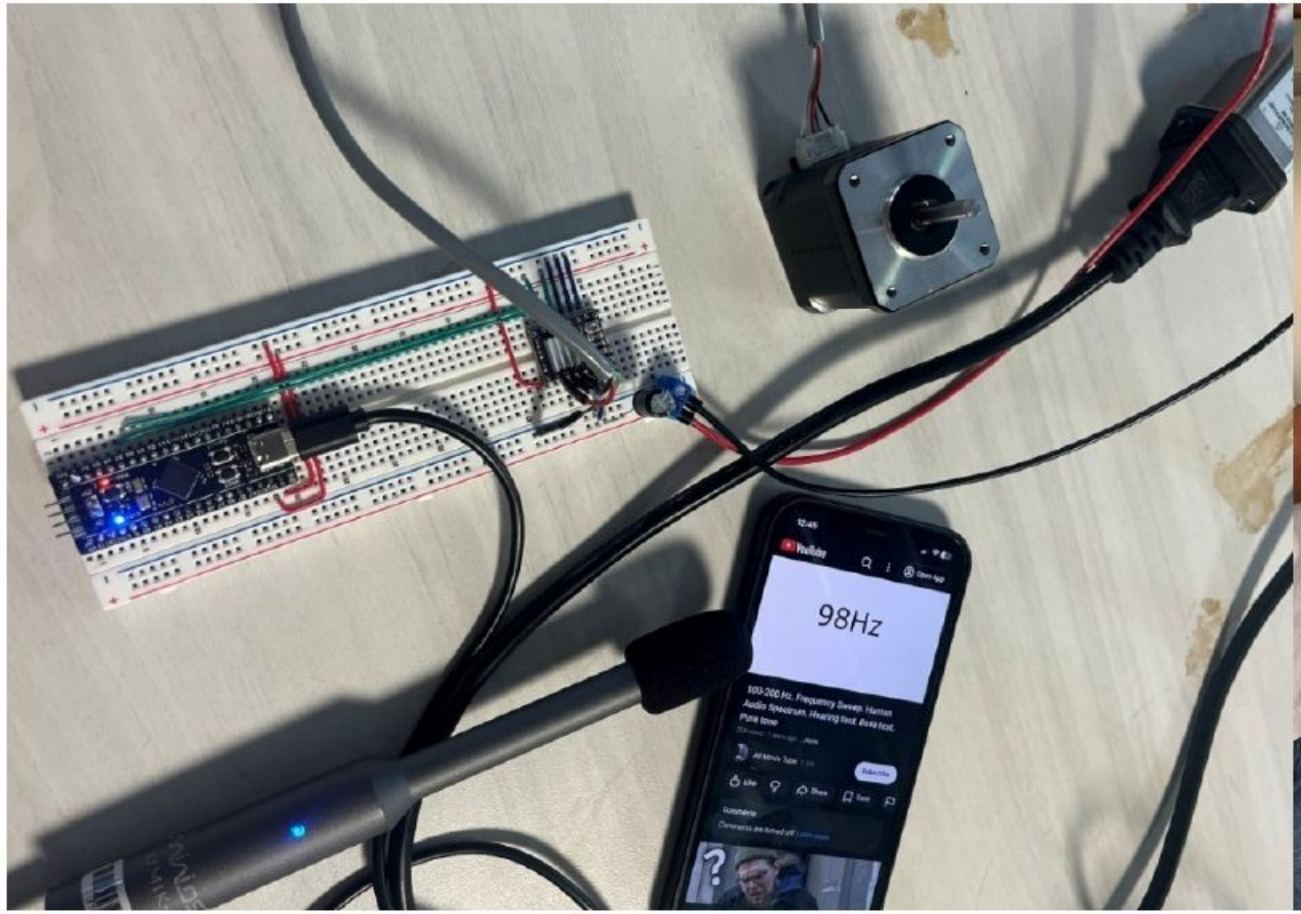
# Appendix F. Physical Verification Prototype CAD

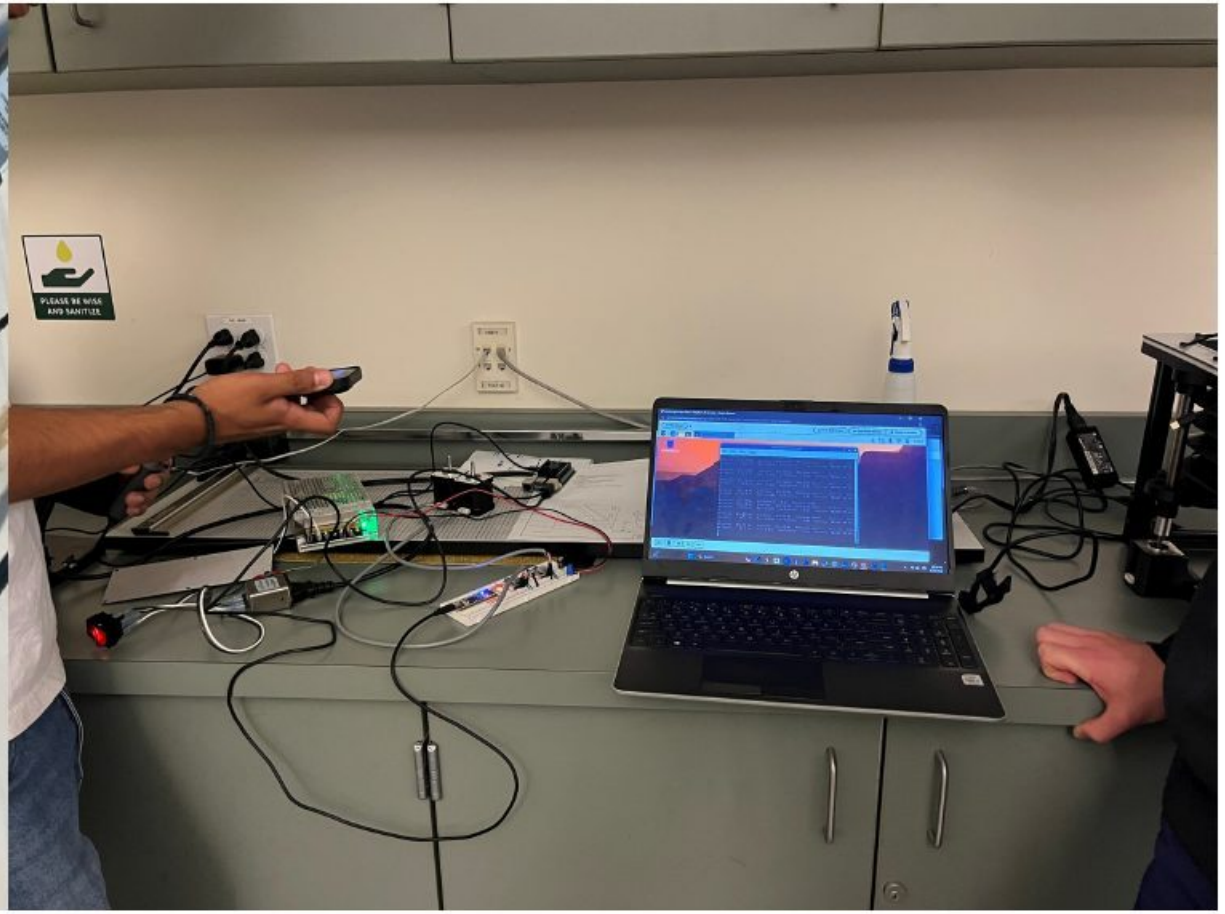





| ITEM NO. | PART NUMBER | QTY. |
|---|---|---|
| 1 | STEPPER MOTOR | 2 |
| 2 | ALUMINUM EXTRUSION | 6 |
| 3 | MOTOR MOUNT | 2 |
| 4 | LEAD SCREW | 2 |
| 5 | LEAD SCREW THREADED NUT | 2 |
| 6 | TOP MOUNT PLATE | 1 |
| 7 | ACCORDIAN STRUCTURE | 1 |
| 8 | THIN MOUNTING PLATE | 1 |
| 9 | LINEAR RAIL | 4 |
| 10 | LINEAR RAIL SLIDER | 4 |
| 11 | BOTTOM PLATE | 1 |
| 12 | TPU SEAL | 1 |
| 13 | BOTTOM ACCORDIAN MOUNTING PIECE | 1 |
| 14 | ALUMINUM MOUNTING PLATE | 1 |
| 15 | RIGHT RAIL MOUNT | 2 |
| 16 | LEFT RAIL MOUNT | 2 |
| 17 | MIC MOUNT | 1 |
| 18 | STEEL BALL BEARING | 2 |
| 19 | MEMBRANE | 1 |

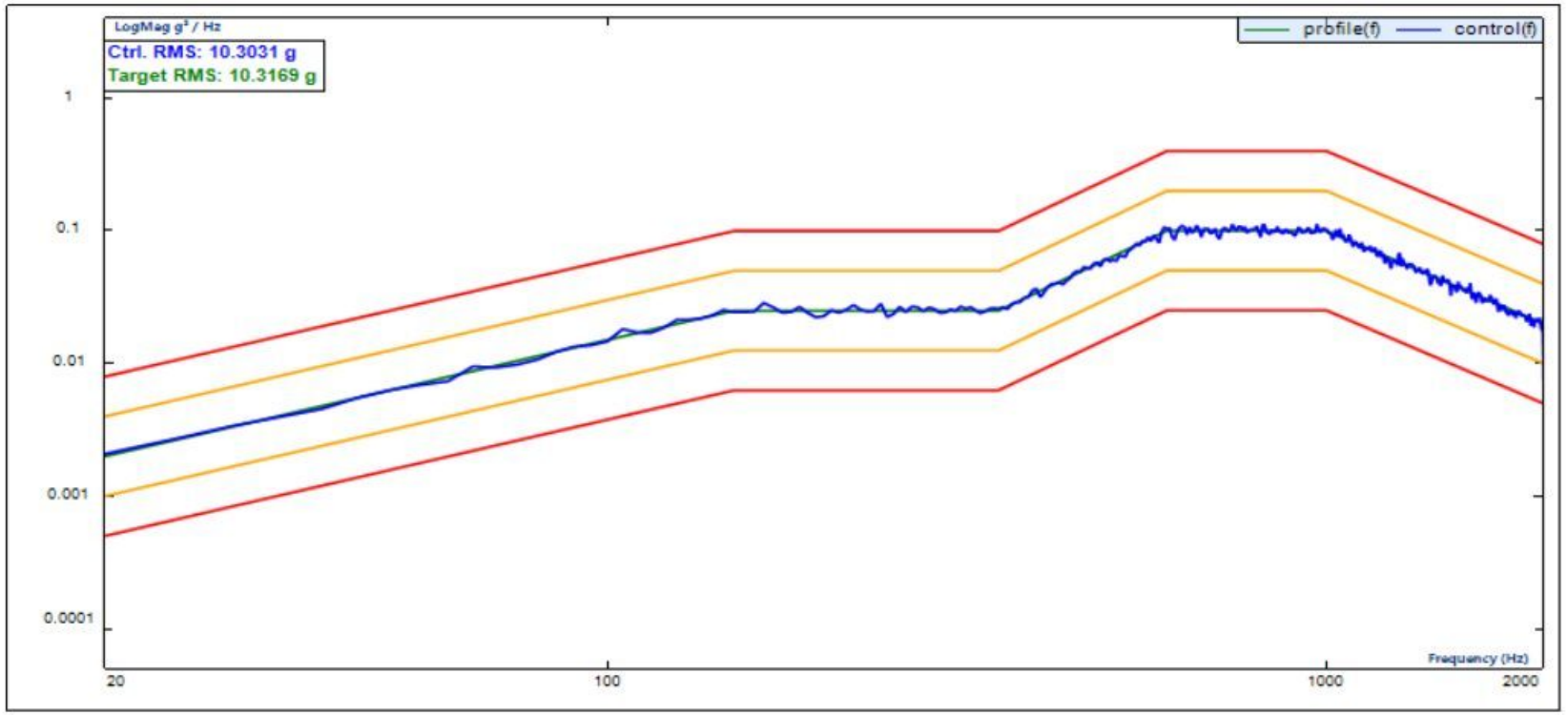
# Appendix G. Prototype Testing Photos

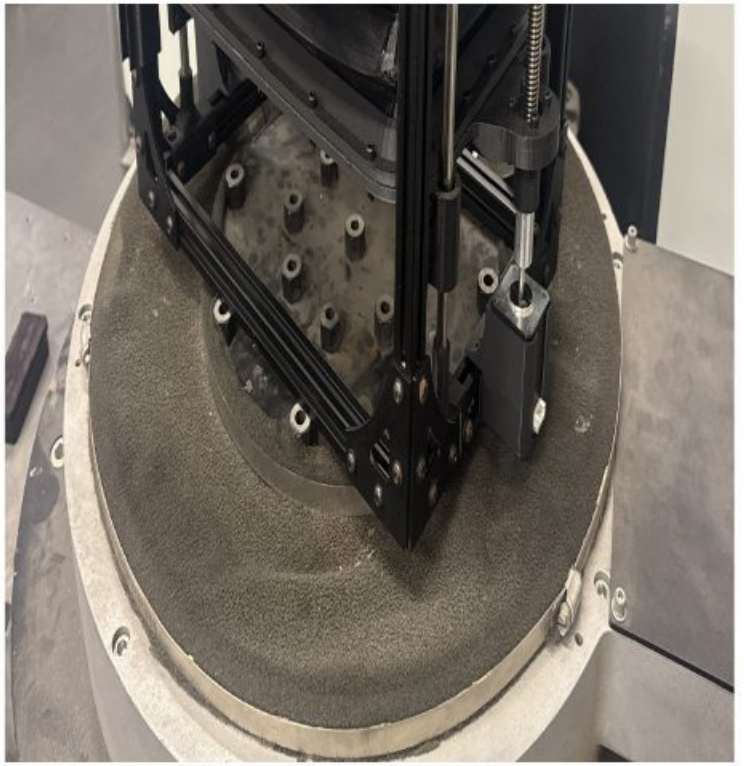


**Frequency Response Verification**

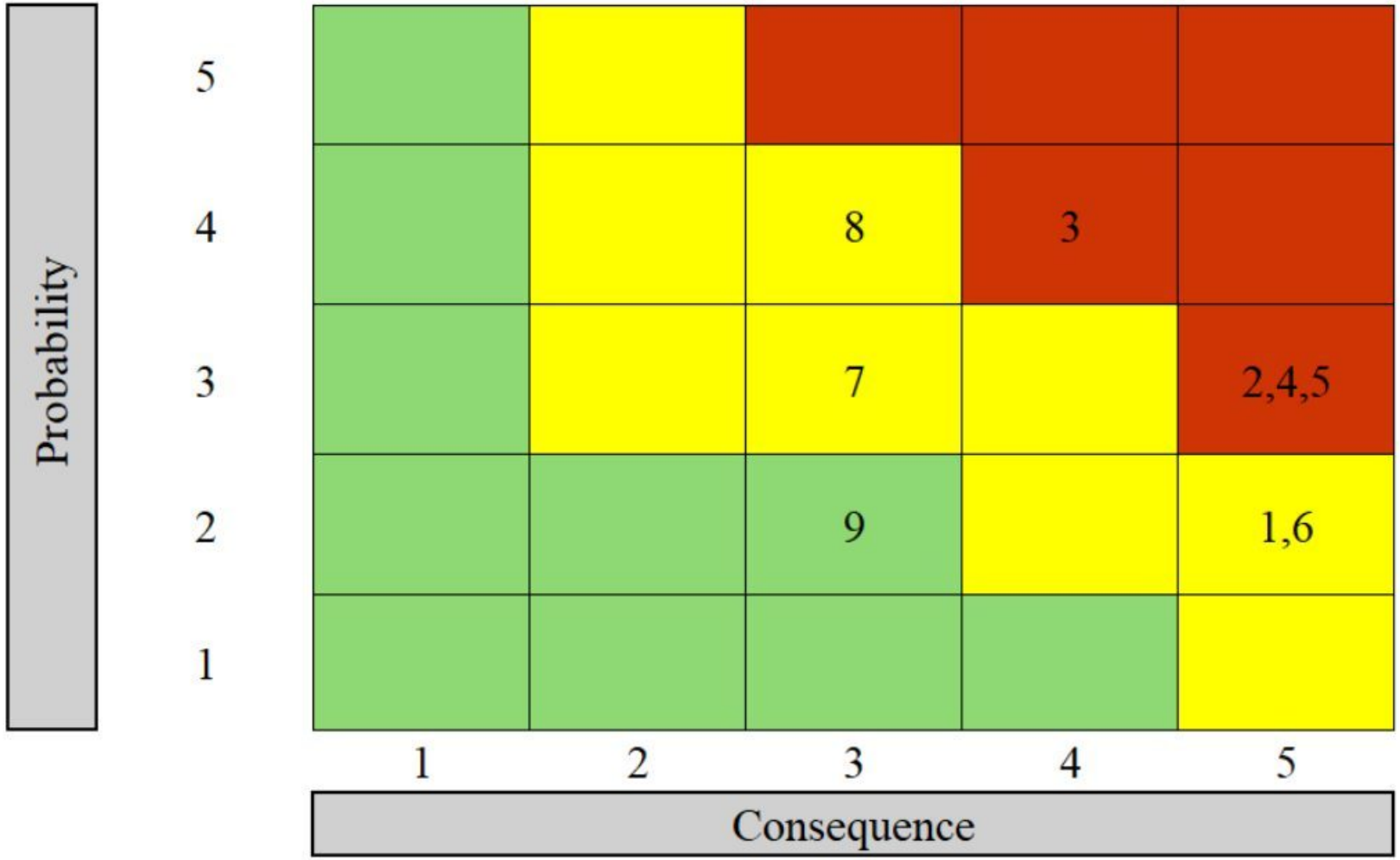
# Appendix G. Prototype Testing Photos Cont.



**Structural Stability Verification**

# Appendix H. Full Risk Matrix



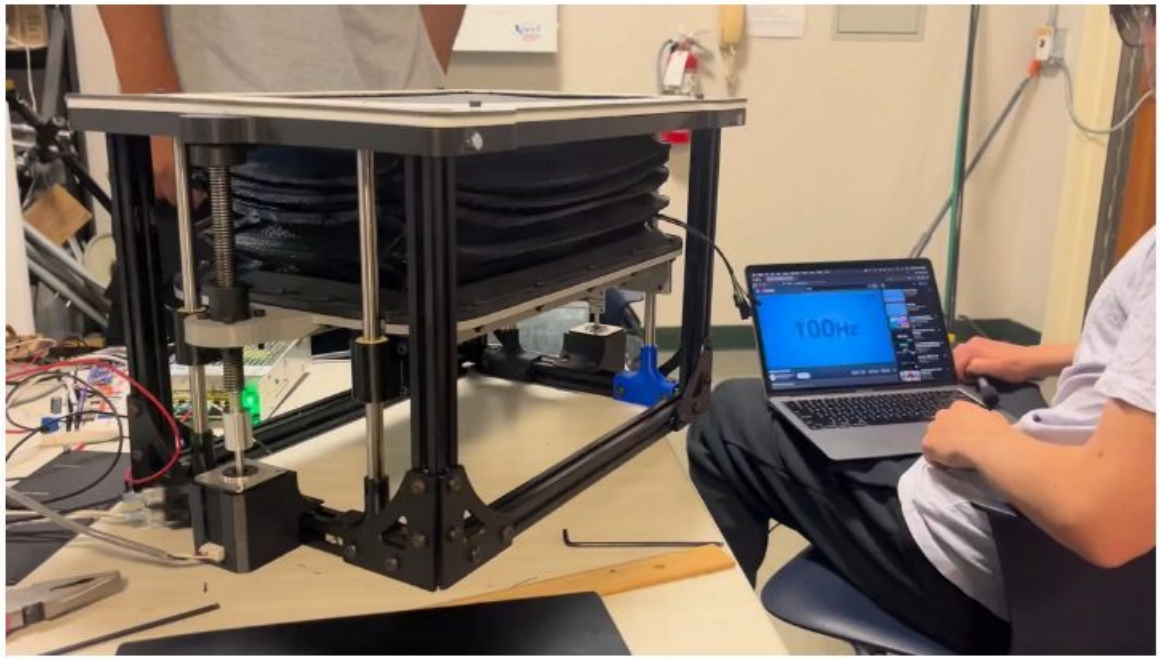
# Appendix H. Full Risk Matrix Cont.

| ID | Risk Name | Description | Related Systems | P | C | Plan |
|---|---|---|---|---|---|---|
| 1 | Membrane Resonation Failure | Membrane fails to resonate at target frequency | Mechanical, HLS | 2 | 5 | Complete analysis and testing of membranes before integration. |
| 2 | Electrical System Failure | Electrical components stop functioning causing linear actuators to no longer move eliminating noise dampening adjustability | Electrical, Control | 3 | 5 | Use electrical and wiring components rated for space, design fail-safe mode where MASS remains fixed at default chamber depth for dominant frequency found in lander during testing |
| 3 | High Pressure Differential | Pressure within the internal reverberation chamber is largely different from the surrounding external pressure | Mechanical, Control | 4 | 4 | Include a controlled pressure equalization aperture, validate sealing behavior during chamber adjustment, emphasize aperture inspection during maintenance |
| 4 | Actuator Positioning Error | Linear actuators move the mounting plate to the incorrect chamber depth | Mechanical, Control | 3 | 5 | Add actuator position feed back, do calibration checks, implement software limits to prevent movement outside allowable depth range |
| **5** | **Acoustic Monitoring Inaccuracy** | **Acoustic sensors inaccurately identify the dominant frequency within the lander** | **Electrical, Control** | **3** | **5** | **Implement extensive sensor calibration, frequency filtering, and acoustic measurements before actuator movement; extensive acoustic testing and programming** |
| 6 | Mounting Failure | Device detaches, loosens, or vibrates from its mounted location during launch or operation | Mechanical, HLS | 2 | 5 | Design mounting brackets for launch loads, use vibration resistant fasteners, perform structural and vibrational testing before system integration |
| 7 | Excessive Mass | Device mass exceeds critical point where it interferes with launch loads | HLS, Propulsion | 3 | 3 | Track device mass throughout design, use lightweight materials, reduce nonessential structural components |
| 8 | Excessive Volume | Device volume takes up excessive space within the HLS interfereing with astronaut living and working space | HLS | 4 | 3 | Minimize protrusion into crew space through optimization of cavity depth and material selection |
| 9 | Power Consumption | System requires excessive power for sensors, controls, and actuation | Electrical, HLS | 2 | 3 | Limit the duty cycle of the actuators, use low power sensors, ensure it is only operational when frequency retuning is required |

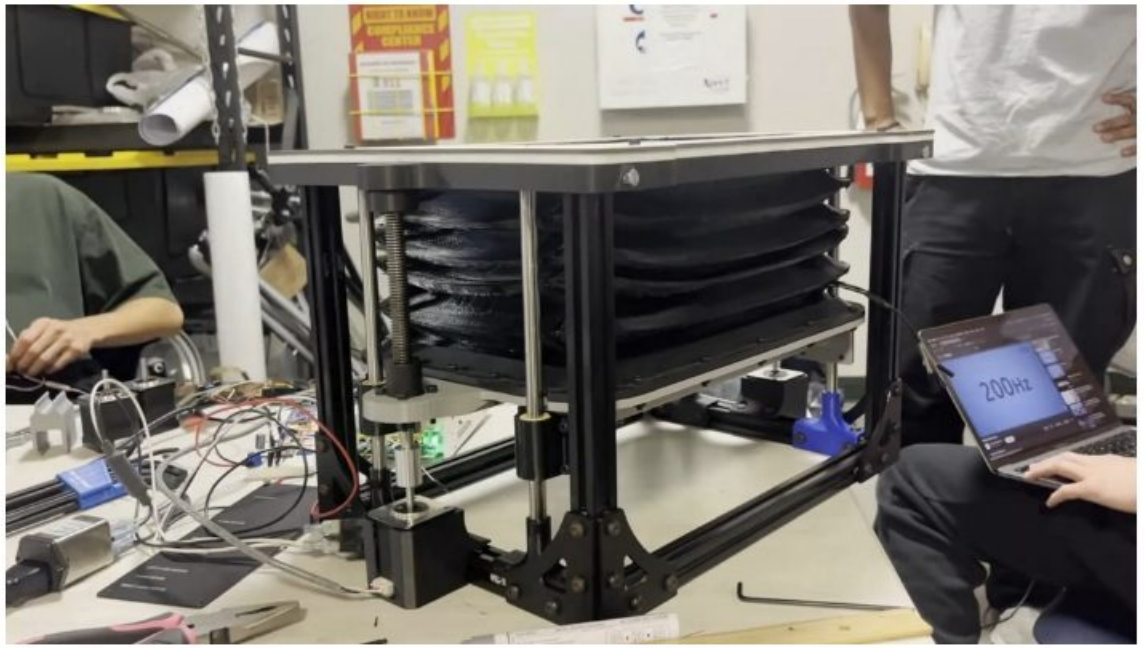
# Appendix I. Full Budget Breakdown

| Mission Phase | Year 1: Dev Phase (2026) | Year 2: Test Phase (2027) | Year 3: Fab & Impl. (2028) | Total ($) |
|---|---|---|---|---|
| **PERSONNEL** | | | | |
| Engineering & Design | 900,000 | 600,000 | 500,000 | 2,000,000 |
| Fabrication & Assembly | 250,000 | 160,000 | 360,000 | 77,0000 |
| Testing & Inspection | 110,000 | 500,000 | 400,000 | 1,010,000 |
| Total Personnel(12) | 1,260,000 | 1,260,000 | 1,260,000 | 3,780,000 |
| **DIRECT COSTS** | | | | |
| Prototyping Hardware | 100,000 | 50,000 | 0 | 150,000 |
| Testing Facilities & Equip. | 0 | 150,000 | 100,000 | 250,000 |
| Final Hardware (1 Unit) | 0 | 0 | 150,000 | 150,000 |
| Total Direct Cost | 100,000 | 200,000 | 250,000 | 550,000 |
| FINAL COST CALC. | | | | |
| ***Total Projected Cost*** | 1,360,000 | 1,460,000 | 1,510,000 | 4,330,000 |
| ***Total Cost Margin(-30%)*** | 408,000 | 438,000 | 453,000 | 1,299,000 |
| **TOTAL PROJECT COST** | **1,768,000** | **1,898,000** | **1,963,000** | **5,629,000** |

# Appendix J. Prototype in Action

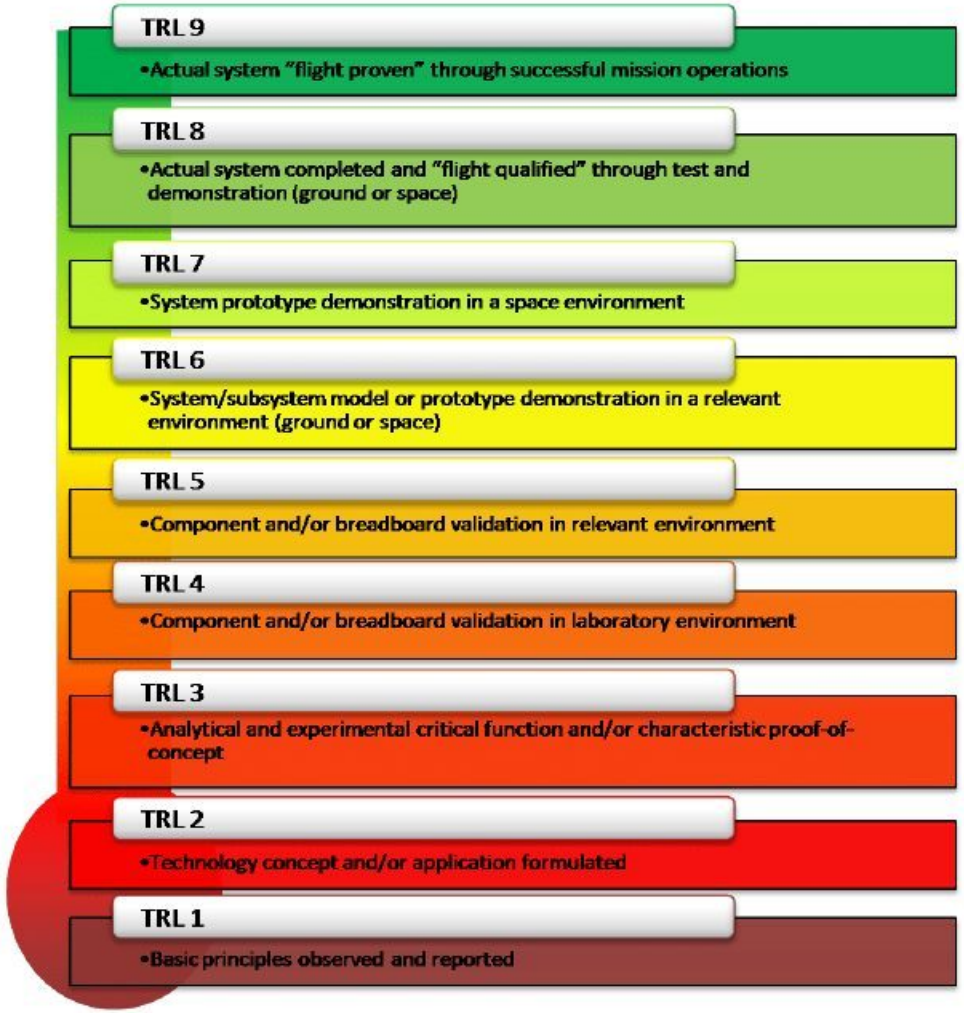



100 Hz Response




200 Hz Response

# Appendix K. Technology Readiness Level Assessment



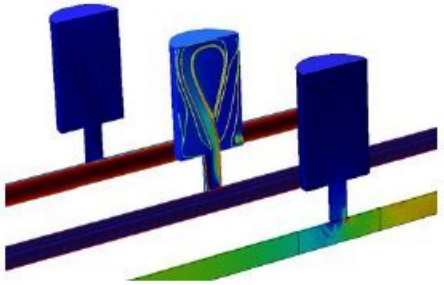

| Evidence | TRL |
|---|---|
| Theory | 2 |
| Simulation | 3 |
| Functional Prototype | 4 |
| Acoustic Chamber Testing | 3 |
| Relevant Environment Testing | 4 |

**Current MASS maturity: TRL 3**

# Appendix L. Why Membrane Absorber Instead of Helmholtz Resonator?

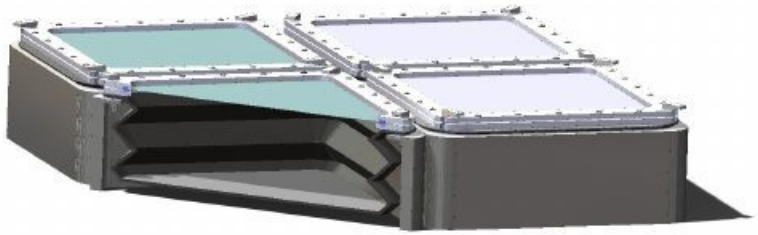
## Helmholtz Resonator

- ✗ Fixed geometry
- ✗ Single tuned frequency
- ✗ Requires multiple resonators
- ✗ Less adaptable



## MASS

- ✓ Adjustable geometry
- ✓ 60–200 Hz range
- ✓ One adjustable unit
- ✓ Adaptive



**MASS was selected because it can adapt to changing ECLSS frequencies in real time.**

# Appendix M. Expected Noise Reduction

**Absorption From Impedance**

$$\alpha = 1 - |R|^2$$

$$R = \frac{z-1}{z+1}$$

$$z = \frac{Z_s}{Z_0}$$

**α** = absorption coefficient (-)
**R** = reflection coefficient (-)
**Zs** = surface impedance of MASS $(Pa \cdot s/m)$
**Z0** = characteristic impedance of air $(Pa \cdot s/m)$

At resonance, MASS is designed so the reactive terms cancel, increasing absorption at the tuned frequency

**Total Cabin Absorption**

$$A_{total} = A_{cabin} + \alpha A_{MASS}$$

where:

$$A_{MASS} = NA_{panel}$$

So more wall coverage means larger acoustic absorption area.

**Estimated SPL Reduction**

$$\Delta L_p = 10log_{10}\left(\frac{A_{after}}{A_{before}}\right)$$

or:

$$\Delta L_p = 10log_{10}\left(\frac{A_{cabin} + \alpha A_{MASS}}{A_{cabin}}\right)$$

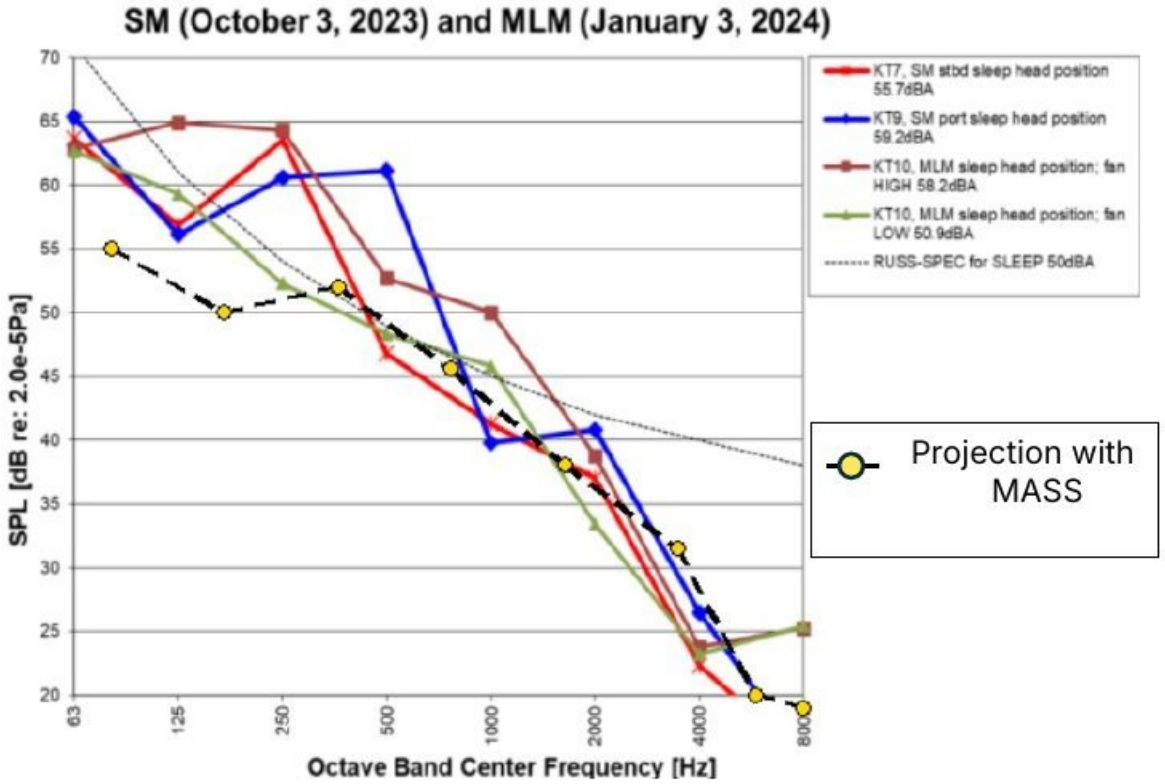
# Appendix M. Expected Noise Reduction

If MASS panels significantly increase total cabin absorption, the expected reduction is:

$$6–10\ dB\ overall\ cabin\ SPL$$

with higher reduction near the tuned resonance frequency.

**More panel area + higher absorption coefficient = greater dB reduction.**
**Final values require acoustic chamber insertion-loss testing.**



# Appendix N. Mounting

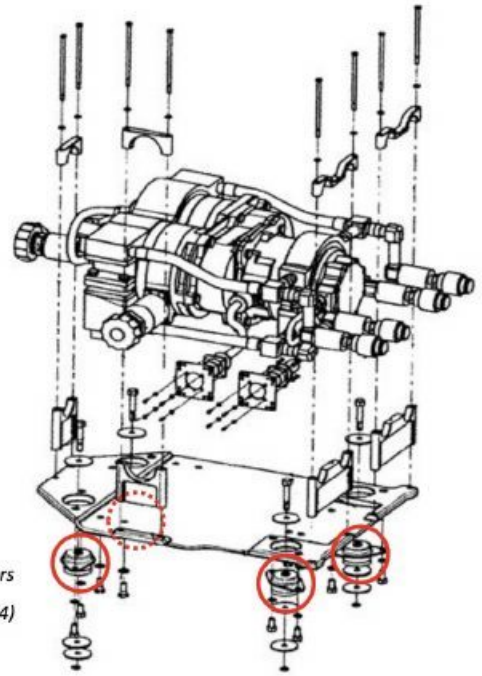



*Figure 35. Locations of the four commercially available isolators used to quiet the U.S. Airlock depressurization pump.*

**Properties**
- ✓ Flexible
- ✓ Durable
- ✓ Appropriate density range

Enables both the membrane function and the accordion chamber function.

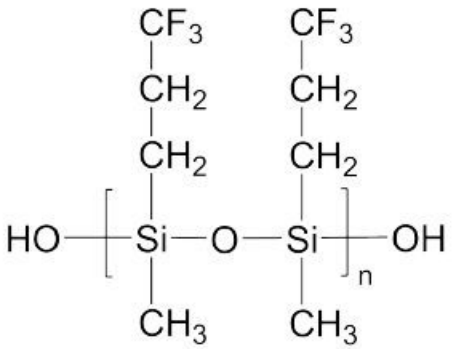


[8]

# Appendix O. Why Fluorosilicone?

**Properties**

- ✓ Flexible (40-80 Shore A)
- ✓ Durable
- ✓ Aerospace heritage
- ✓ Maintains pressure seal
- ✓ Appropriate density range
- ✓ Chemical resistance
- ✓ Temperature stability(-60°C to 204°C)
- ✓ Used in industry for jet fuel system






**Fluorosilicone O-rings**

**Enables both the membrane function and the accordion chamber function**

# Appendix P. Timeline - Quarterly

